

Does a Public VizWiz Checkpoint Improve Selective Reliability?

A Risk-Controlled Audit of Assistive VQA

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Abstract

Vision-language models can become more accurate after domain fine-tuning while remaining unreliable about when to answer. We audit a base model and a public checkpoint presented as VizWiz-tuned under a deliberately narrow safety definition: avoiding *entirely wrong* answers while retaining useful coverage. Questions are split once, by a simple seeded random assignment, into a threshold and calibrator-fit set (20%), an independent certification set (20%), and an untouched test set (60%). The primary selection score is mean answer log-likelihood, and the primary error event is an official VQA consensus score of zero. The confirmatory family contains the two natural-regime likelihood policies. Each fit-set threshold is checked exactly once on certification using a one-sided Clopper-Pearson upper bound at $\alpha = 0.10$ and Bonferroni-adjusted $\delta = 0.025$ per policy, controlling the probability of any false certification in the two-policy family at most 0.05. Thresholds are then frozen for test evaluation. This control is conditional on i.i.d. sampling and both checkpoints being independent of the certification labels. On 2581 untouched natural questions, the public tuned checkpoint raises mean VQA from 69.5% to 72.6% and lowers the entirely-wrong rate from 20.5% to 16.6%. Neither policy’s certification bound (15.71% base; 41.28% tuned) meets the 0.10 target, so prescribed deployed coverage is 0.0% for base and 0.0% for tuned. This audit concerns selective factual reliability on one assistive-VQA distribution; it does not establish jailbreak safety, fairness, privacy, robustness to arbitrary shift, or user benefit.

1 Introduction

Visual question answering (VQA) is a plausible interface for assisting blind and low-vision users, but wrong answers can be more costly than explicit uncertainty. VizWiz captures this deployment pressure through naturally posed questions, imperfect mobile images, conversational language, and many questions that cannot be answered from the image [1]. Image quality and missing content are distinct causes of failure [2], while long-form answers can sound plausible even when they hallucinate details on unanswerable inputs [3].

Domain fine-tuning is therefore ambiguous from a reliability perspective. It may increase average task accuracy, but it may also reshape confidence rankings, overfit domain regularities, or erode behavior inherited from the base model. Vision-language safety fine-tuning studies show that post-training can change safety behavior in either direction [4]; that broader notion of harmful-content safety is outside our endpoint. Here, *safer* means only that a system avoids entirely wrong VQA answers at a prespecified risk tolerance while answering as many questions as possible.

Selective VQA makes this trade-off explicit through risk and coverage [5, 6]. This study contributes:

- a paired base-versus-domain-tuned audit using identical questions, prompts, decoding, scoring, and split assignments;
- an independent fit/certify/test protocol that prevents test-time threshold tuning and reports an exact one-sided certification decision; and
- exploratory grounding and evidence-acquisition diagnostics that are kept separate from the primary certificate.

The central question is not whether a public tuned checkpoint improves a leaderboard score, but whether it improves the attainable coverage of answers that survive a fixed, auditable risk gate.

2 Related Work

Assistive VQA and visual evidence. VizWiz was collected from blind users in a natural VQA setting [1]. Subsequent work decomposes unanswerability into quality failures and missing content [2], and shows that VQA systems can answer without localizing the evidence used by human annotators [7]. These findings motivate abstention, but also suggest more useful follow-up actions such as retaking, reframing, or clarifying an image.

Selective VQA and calibration. Reliable VQA evaluates the error among answered questions against the fraction answered [5]. Learned selectors improve this trade-off, yet confidence thresholds can degrade under mixtures of in-distribution and out-of-distribution inputs [6]. For generative multi-modal models, likelihood- and sampling-based confidence estimators behave differently, with no universal winner [8]. More broadly, post-hoc calibration can deteriorate under dataset shift [9]; we therefore distinguish in-distribution certification from exploratory shift results.

Risk control. Learn-then-test methods calibrate fixed predictive procedures using hypothesis tests [10]. Conformal risk control extends finite-sample control to bounded monotone losses [11]. Weighted conformal methods require additional assumptions under covariate shift [12]. Dedicated methods for controlling risk among selected predictions are still developing [13]. Our primary analysis is intentionally simpler: a fit-only threshold is subjected to one independent exact binomial certification check, and the resulting threshold and decision are never revised using test data.

3 Audit Protocol

3.1 Models and inference

We evaluate two pinned revisions. The base is `Qwen3-VL-8B` at commit `0c351dd`; the tuned model is `Qwen-VizWiz-8B` at commit `984042c`. The tuned checkpoint’s public model card identifies an Unsloth 4-bit derivative of Qwen3-VL as its starting point and names Unsloth and TRL as training tools, but does not document its corpus, optimizer, update count, checkpoint-selection rule, or whether the VizWiz validation split was excluded. We therefore audit two public pipelines and do not treat their difference as an identified causal effect of fine-tuning.

Both checkpoints are loaded as `bfloat16` with Transformers 4.57.6 and PyTorch 2.6 on a single NVIDIA A10. Images are capped at 1,024 visual tokens. Both pipelines receive the same image and question and the same instruction: answer from visible evidence only, emit exactly “unanswerable” when evidence is insufficient, and otherwise return a short answer without explanation. Decoding is greedy, batch size is one, and the output limit is 32 tokens. No calibration label is available to either model during generation. For deterministic storage, every decoded natural image is converted to RGB and re-encoded once as JPEG at quality 95; both models receive the same stored file.

Answers are normalized and scored against the human references using the standard VQA leave-one-out consensus rule. Mean VQA score is a secondary utility measure; the primary binary loss treats partial credit differently from a complete miss:

$$E_i = \mathbb{1}\{\text{VQA}(\hat{y}_i, Y_i) = 0\}. \quad (1)$$

Thus $E_i = 1$ only when the generated answer receives no annotator-supported credit. The broader semantic mapping used to identify explicit abstentions is not used to rewrite predictions for this official score.

3.2 Primary selection score

For the T_i non-special tokens in the generated answer, the primary score is the length-normalized conditional log-likelihood

$$S_i = \frac{1}{T_i} \sum_{t=1}^{T_i} \log p_{\theta}(\hat{y}_{it} \mid x_i, q_i, \hat{y}_{i,<t}). \quad (2)$$

Higher values indicate greater confidence. Let $G_i = 1$ only when the decoded response contains at least one non-special token and is not semantically mapped to the explicit abstention label “unanswerable.” A threshold τ yields the frozen substantive-answer policy $A_i(\tau) = G_i \mathbb{1}\{S_i \geq \tau\}$. Coverage and selective risk on a set D are

$$\hat{C}_D(\tau) = \frac{1}{|D|} \sum_{i \in D} A_i(\tau), \quad \hat{R}_D(\tau) = \frac{\sum_{i \in D} A_i(\tau) E_i}{\sum_{i \in D} A_i(\tau)}. \quad (3)$$

We report the risk as undefined when no substantive answer is issued; zero coverage is not counted as a successful certificate. Thus a confident “unanswerable” may be a useful semantic abstention, but it cannot inflate answer coverage.

3.3 Frozen split and threshold fitting

We load all 4,319 examples in the `val` split of `lmms-lab/VizWiz-VQA`, pinned at commit `d428a2d`. Eighteen source questions used to validate instrumentation are frozen as a disjoint pilot and excluded from every analysis split, leaving 4,301 evaluation questions. With question as the sampling unit, a single pseudo-random permutation using seed 20260712 assigns 20% to fit F , 20% to independent certification C , and 60% to untouched test T . The assignment is shared by both models and stored before final inference. This gives 860 questions in F , 860 in C , and 2581 in T . There is no stratification, resampling, or threshold-specific split.

For each model separately, we enumerate score cutpoints on F without splitting ties and choose the maximum-coverage candidate satisfying $\hat{R}_F(\tau) \leq 0.05 = \alpha/2$. Ties between equally covering policies use the stricter threshold. If no nonempty candidate exists, the policy abstains on all questions and cannot pass certification. The $\alpha/2$ margin is only a conservative power heuristic: this search is fit-set model selection, and independent certification supplies validity.

3.4 One-shot certification and untouched test

Let $n_C = \sum_{i \in C} A_i(\hat{\tau})$ and $e_C = \sum_{i \in C} A_i(\hat{\tau}) E_i$. For $0 \leq e_C < n_C$, the one-sided exact upper bound is

$$U_{CP}(e_C, n_C; \delta) = \text{Beta}^{-1}(1 - \delta; e_C + 1, n_C - e_C). \quad (4)$$

We set the bound to one when $n_C = 0$ or $e_C = n_C$. The confirmatory family is fixed to the base and fine-tuned natural-regime likelihood policies. A pipeline passes exactly when $U_{CP} \leq \alpha$, with $\alpha = 0.10$ and $\delta = 0.025$ per policy. By Bonferroni’s inequality, the probability of any false certification in this two-policy family is at most 0.05 under i.i.d. question sampling from the target distribution, provided both frozen checkpoints are independent of the certification labels. Equivalently, conditional on the frozen fit-stage pipelines, the probability over the certification draw that either policy is deployed with population $\Pr(E = 1 \mid A = 1) > 0.10$ is at most 0.05. This is a repeated-sampling false-certification guarantee, not a posterior probability that a policy is safe after it passes. Certification is queried once

per frozen primary pipeline: we do not inspect alternative thresholds on C , retune after failure, or use C for Platt fitting.

No complete pretraining-data audit is available for either checkpoint, so label independence is an explicit condition for both. The concern is especially acute for the tuned checkpoint because its public provenance does not establish exclusion of VizWiz validation labels.

After this decision, $\hat{\tau}$ is applied unchanged to T . We retain its candidate coverage, selective risk, and accepted/error counts whether certification passes or fails, but headline deployed metrics use abstain-all after failure, and only a passing pipeline supports the prespecified risk-control claim. The test set is used once for the final comparison and never to modify either policy. The Bonferroni statement applies only to the two confirmatory policies. Any secondary grounding, shift, or acquisition check uses marginal $\delta = 0.05$ and is not a simultaneous guarantee. Operationally, a failed primary policy defaults to abstain-all; its frozen-threshold test metrics are retained only as diagnostic counterfactuals.

3.5 Calibration and exploratory diagnostics

Calibration is descriptive and does not alter the primary gate. A two-parameter Platt map $g(s) = \sigma(as+b)$ is fit on standardized scores from F only, with an L2 slope penalty of 10^{-4} , to predict $\mathbb{1}\{E_i = 0\}$. Its parameters are frozen before evaluating 10-bin expected calibration error (ECE), Brier score, and reliability diagrams on T [14]. If F contains only one correctness class or numerical optimization fails, an add-one-half (Jeffreys-prior mean) constant predictor is used and recorded as a fallback; neither C nor T is used for fitting.

Two additional score families are exploratory. A visual-sensitivity score contrasts the exact generated token sequence’s mean log-likelihood under the real image with its teacher-forced likelihood under a same-size mid-gray image. This is not evidence of semantic grounding. We also select 300 answerable fit, 300 answerable certification, and 600 answerable test questions before inference, balanced across Gaussian blur (radius 4), 25% brightness, and downsampling to 96 pixels on the longest side followed by upsampling. The answer is generated once from the corrupted image. An acquisition score is the maximum of its corrupted- and clear-image visual-sensitivity scores; the model is not allowed to regenerate after seeing the clear image. We show per-pipeline risk-coverage curves and failure examples, but do not reuse the primary certificate or claim simultaneous guarantees. Any exact bounds for these diagnostics are exploratory marginal checks at $\delta = 0.05$. Each exploratory model-regime-score combination repeats the same fit-set threshold rule, one-shot certification check, and test freeze defined above; a failed check deploys abstain-all. Each uses the corresponding $F/C/T$ split, and no natural-regime threshold or certificate is transferred to corruption.

4 Results

4.1 Primary outcome

Table 1 records the only confirmatory certification checks. Neither natural-regime policy meets the numerical gate: the base and tuned CP upper bounds are 15.71% and 41.28%, both above 0.10. The base certification and deployed-test counts are 17/168 and 0/0 errors/accepted; the tuned counts are 1/11 and 0/0. The failed frozen thresholds would have accepted 32/420 and 0/35 errors/answers on test, respectively (16.3% coverage at 7.6% empirical risk and 1.4% at 0.0%). These candidate outcomes are diagnostics, not certificates; in particular, zero errors among 35 tuned answers cannot repair a bound learned from only 11 accepted certification examples. Conditional on the frozen certification outcomes, the paired difference in audit-policy test coverage (tuned minus base) is 0.0 pp with a 95% percentile-bootstrap interval [0.0 pp, 0.0 pp]. This degenerate interval is a mechanical consequence of both policies failing; it does not show equivalence. The tuned checkpoint is described as improving selective reliability

Table 1: Primary audit populated from the frozen analysis artifact. “CP upper” is the one-sided certification bound at $\delta = 0.025$ in Eq. 4; decisions use unrounded bounds and test risk is descriptive for the already frozen policy. A pass is marked * because it is conditional on no evaluation-label overlap.

Pipeline	Test mean VQA	$\hat{\tau}$	Cert. e/n	Cert. cov.	CP upper	Pass?	Test cov.	Test risk	ECE / Brier
Base	69.5%	-0.002	17/168	19.5%	15.71%	Fail	0.0%	–	0.033 / 0.148
Public tuned	72.6%	-0.022	1/11	1.3%	41.28%	Fail	0.0%	–	0.061 / 0.128

only if its frozen policy provides more useful coverage at the target risk without relying on a failed or post-hoc certificate.

Figure 1 gives eligible-only normalized partial-AURC values of 0.151 for the base pipeline and 0.131 for the tuned pipeline (lower is better). Each is normalized by its own maximum substantive-answer coverage (59.4% and 46.0%), so AURC is a descriptive ranking diagnostic rather than an overall reliability comparison. The unrestricted paired VQA-score difference is 3.1 pp with a 2,000-resample percentile paired-bootstrap interval [2.0 pp, 4.3 pp] at seed 20260712. The entirely-wrong rate changes from 20.5% to 16.6% (paired difference -3.9 pp, interval [-5.2 pp, -2.6 pp]), while the substantive-answer rate changes by -13.4 pp [-14.8 pp, -12.1 pp]. Thus the public tuned checkpoint has better average task utility and fewer completely wrong outputs, but it also declines to answer much more often; neither fact supplies the missing selective-reliability certificate.

4.2 Calibration and shift

Platt-scaled ECE and Brier scores are 0.033 and 0.148 for the base pipeline and 0.061 and 0.128 for the public tuned pipeline. These scalar summaries are descriptive probability-quality checks; they do not replace the exact certification decision. They are also mixed: ECE and correctness AUROC (0.798 versus 0.760) favor the base pipeline, whereas Brier score and normalized partial AURC favor the tuned pipeline.

On the balanced answerable corruption test ($n = 600$), the separately fitted base likelihood policy passes its exploratory marginal check with certification $e/n = 3/82$ and a 9.18% upper bound; it then records 10/154 errors/accepted answers at 25.7% coverage and 6.5% observed risk. The tuned policy’s certification $e/n = 0/6$ has a 39.30% upper bound and fails, so its prescribed coverage is 0.0% (the failed candidate records 0/7 errors/accepted answers on test). Aggregate VQA is 57.8% versus 56.1% (paired difference -1.7 pp, interval [-4.0 pp, 0.5 pp]); entirely-wrong rates are 26.8% and 26.5% (difference -0.3 pp, interval [-3.0 pp, 2.2 pp]). The tuned model’s substantive-answer rate is also lower (41.0% versus 57.3%). Low resolution has the highest entirely-wrong rate for both pipelines. This study is exploratory, marginal, answerable-only evidence about failure behavior, not a transferred or family-wise robustness guarantee.

4.3 Grounding and acquisition

The mid-gray-image contrast worsens natural-data correctness ranking: base AUROC falls from 0.798 to 0.399 and normalized partial AURC rises from 0.151 to 0.237; the corresponding tuned values move from 0.760 to 0.494 and 0.131 to 0.182. On corrupted images, clear-image acquisition produces the same certification and accepted/error counts as the visual-delta score for each model, with only small and inconsistent ranking changes. Because the answer is frozen before acquisition, it cannot turn an entirely wrong response into a correct one here. These diagnostics do not support a grounding or acquisition benefit; an acquire-and-regenerate policy requires a separate experiment.

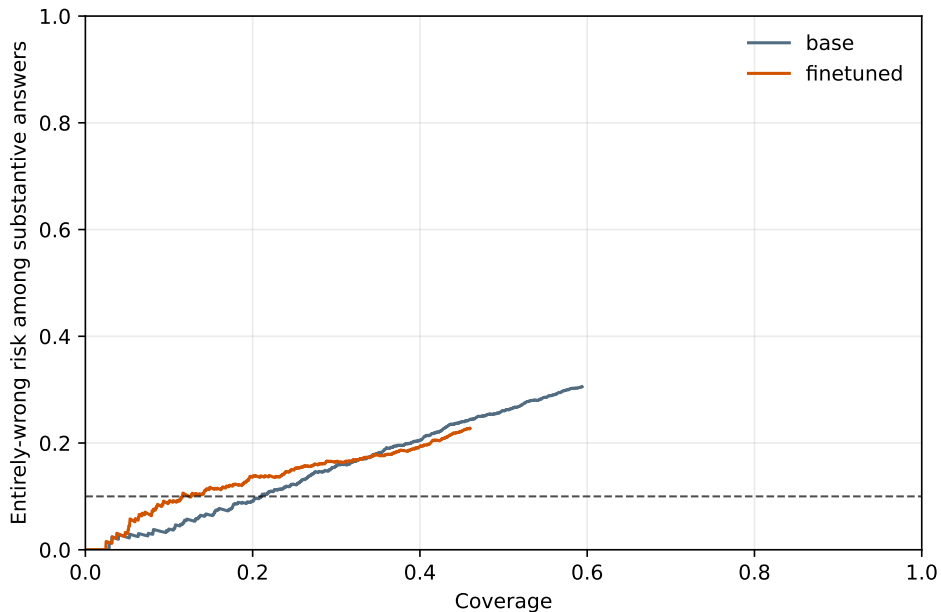


Figure 1: Test risk-coverage curves for the primary score. Curves are descriptive and include only substantive answers; filled markers denote audit operating points that met the numerical certification gate.

5 Failure Analysis

We conduct a Codex-assisted case-by-case review of the 15 highest-confidence substantive-answer test errors in each model-by-regime stratum, ordered by score then case ID, and assign one primary category: unrecognizable image, target outside the frame, OCR failure, question ambiguity, unsupported object/detail, or reference disagreement. Review is conducted after primary metrics are frozen and cannot change correctness labels. For publication we take the first privacy-safe unique source from each of the four strata; uncertain or identifying images are excluded without replacement by reviewer preference.

Across 60 reviewed model-by-regime rows (52 unique case IDs and 48 unique source questions), 39 are reference disagreements, nine OCR failures, five question ambiguities, three unsupported-detail errors, two unrecognizable images, and two targets outside the frame. The base candidate selector misses all 15 natural cases and 10/15 shifted cases; the tuned candidate selector catches all 30 reviewed cases. These are deterministic top-score audit counts, not prevalence estimates. All four displayed rows pass the prespecified privacy screen; their source fields are retained in `publication_examples.csv`, while the full labels and exclusions are retained in `failure_review.csv`. For natural data, “candidate” remains counterfactual because the failed certification gate forces the deployed policy to abstain on every case.

This review distinguishes a confidence-ranking failure from a perception failure: a correct abstention on missing evidence is desirable, whereas an answer that happens to match references without using the image is not evidence of grounding. The latter distinction follows the VizWiz grounding literature [7] and is not settled by the primary certificate.

6 Limitations and Claim Boundary

Narrow endpoint. The event E_i captures entirely wrong short answers, not partial misinformation, harmful advice, bias, privacy leakage, or adversarial misuse. Passing therefore supports only the stated selective-reliability claim. No BLV participant study is performed, so the paper does not establish

Table 2: Four stratum-balanced high-confidence errors selected by a rule frozen before inspecting examples.

Case	Question / image condition	Output / official score	Candidate selector behavior	Appropriate action
Base / 1064	Login screen or still installing? Natural image.	“login screen”; VQA = 0; $\ell = -6.96 \times 10^{-6}$.	Above -0.0015 : miss. Image supports a login/welcome reading.	Adjudicate or expand references.
Tuned / 2670	All-purpose or self-rising flour? Type is outside the frame.	“self rising”; VQA = 0; $\ell = -0.0308$.	Below -0.0219 : catch. The answer is unsupported.	Request a front-label image or abstain.
Base / 3106-L	Book-bag color under low light.	“blue”; VQA = 0; $\ell = -2.18 \times 10^{-5}$.	Above -0.0126 : miss. References say blue-grey.	Adjudicate acceptable color granularity.
Tuned / 3456-R	Tin contents at low resolution.	“baked beans and sausages”; VQA = 0; $\ell = -0.0811$.	Below -0.0277 : catch. References omit only “and.”	Normalize conjunctions or adjudicate.

usability, preference, or real-world benefit.

Statistical scope. The exact binomial bound relies on an independently selected threshold and i.i.d. question sampling from the target distribution for the frozen policy. Mere exchangeability is not asserted as sufficient. A simple question-level split does not protect against unidentified near duplicates or future distribution shift. The two confirmatory natural-regime policies use $\delta = 0.025$ each, giving family-wise false-certification probability at most 0.05 by Bonferroni. Secondary slices, corruption studies, grounding scores, and acquisition scores are exploratory marginal analyses and should carry multiplicity-aware uncertainty if used for more than description.

Measurement and causal scope. VQA consensus labels are noisy and can assign partial credit to answers with different practical value. Mean answer log-likelihood is length normalized but still depends on tokenization and decoding. Platt calibration can improve ECE without improving the ordering needed for selective coverage. Finally, a causal claim about *domain fine-tuning* requires the checkpoints to differ only by the documented tuning intervention; otherwise the comparison is between two pipelines, not an isolated treatment effect. The single Codex-assisted qualitative coding pass has no independent second reviewer, so its category counts are diagnostic rather than reliability-tested annotations.

7 Ethics and AI Assistance

The study uses an existing assistive-VQA benchmark and does not infer users’ identities or disability status. The four displayed qualitative examples were screened for private or identifying content before inclusion. Deployment language should preserve user agency: abstention is a request for more evidence or human help, not a judgment about the user’s ability to pose a question.

OpenAI Codex assisted with literature retrieval, protocol refinement, implementation, experimental orchestration, statistical checks, qualitative failure categorization, and manuscript drafting. The author selected the research question and checkpoints, directed the work, and is responsible for source verification, interpretation, and every final numerical claim. No generated value is treated as an experimental result unless it is reproduced from the archived prediction and analysis artifacts.

8 Conclusion

We asked whether a public checkpoint presented as domain-tuned exhibits better selective reliability than its base family under a prespecified entirely-wrong risk target. The tuned checkpoint improves average VQA and reduces entirely-wrong outputs, but neither likelihood policy passes independent certification, so neither provides certified selective coverage at the target risk. This outcome compares two public pipelines and does not identify fine-tuning as its cause. It is bounded to the frozen VizWiz evaluation and the fit/certify/test protocol described above; broader safety and user-impact claims require separate evidence.

References

- [1] Danna Gurari, Qing Li, Abigale J. Stangl, Anhong Guo, Chi Lin, Kristen Grauman, Jiebo Luo, and Jeffrey P. Bigham. VizWiz grand challenge: Answering visual questions from blind people. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 3608–3617, 2018. URL <https://arxiv.org/abs/1802.08218>.
- [2] Tai-Yin Chiu, Yinan Zhao, and Danna Gurari. Assessing image quality issues for real-world problems. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3646–3656, 2020. URL <https://arxiv.org/abs/2003.12511>.
- [3] Mina Huh, Fangyuan Xu, Yi-Hao Peng, Chongyan Chen, Hansika Murugu, Danna Gurari, Eunsol Choi, and Amy Pavel. Long-form answers to visual questions from blind and low vision people. In *Conference on Language Modeling*, 2024. URL <https://arxiv.org/abs/2408.06303>. Oral Spotlight.
- [4] Yongshuo Zong, Ondrej Bohdal, Tingyang Yu, Yongxin Yang, and Timothy Hospedales. Safety fine-tuning at (almost) no cost: A baseline for vision large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 62867–62891. PMLR, 2024. URL <https://proceedings.mlr.press/v235/zong24a.html>.
- [5] Spencer Whitehead, Suzanne Petryk, Vedaad Shakib, Joseph Gonzalez, Trevor Darrell, Anna Rohrbach, and Marcus Rohrbach. Reliable visual question answering: Abstain rather than answer incorrectly. In *Computer Vision – ECCV 2022*, pages 148–166. Springer, 2022. doi: 10.1007/978-3-031-20059-5_9. URL <https://arxiv.org/abs/2204.13631>.
- [6] Corentin Dancette, Spencer Whitehead, Rishabh Maheshwary, Ramakrishna Vedantam, Stefan Scherer, Xinlei Chen, Matthieu Cord, and Marcus Rohrbach. Improving selective visual question answering by learning from your peers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24049–24059, 2023. doi: 10.1109/CVPR52729.2023.02303. URL <https://arxiv.org/abs/2306.08751>.
- [7] Chongyan Chen, Samreen Anjum, and Danna Gurari. Grounding answers for visual questions asked by visually impaired people. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 19098–19107, 2022. URL <https://arxiv.org/abs/2202.01993>.
- [8] Julian Eisenschlos, Hernán Maina, Guido Ivetta, and Luciana Benotti. Selectively answering visual questions. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 4219–4229. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.250. URL <https://aclanthology.org/2024.findings-acl.250/>.
- [9] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://arxiv.org/abs/1906.02530>.
- [10] Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *arXiv preprint arXiv:2110.01052*, 2021. URL <https://arxiv.org/abs/2110.01052>.
- [11] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2208.02814>.
- [12] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://arxiv.org/abs/1904.06019>.
- [13] Tian Bai and Ying Jin. Conformal selective prediction with general risk control. *arXiv preprint arXiv:2603.24704*, 2026. URL <https://arxiv.org/abs/2603.24704>. Preprint.
- [14] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.